Unit 3: Computer Architecture

What you need to know: Computer architecture is all about the physical hardware components inside the computer case and how they connect, communicate, and work together as a system.

- **CPU (Central Processing Unit)**
  - **What it is:** The "brain" of the computer.
  - **Study Tip:** It handles all the processing, calculations, and instructions. Simply put, without the CPU, the computer cannot do anything.
- **ALU (Arithmetic Logic Unit)**
  - **What it is:** One of the core components located inside the CPU.
  - **Study Tip:** As the name suggests, it handles two specific things: **Math** (Arithmetic like addition, multiplication) and **Decision-making** (Logic operations, like the AND/OR/NOT gates you studied in Unit 2).
- **Control Unit**
  - **What it is:** The other main component inside the CPU.
  - **Study Tip:** Think of it as the computer's "Traffic Cop" or "Manager." It doesn't do the actual math; instead, it directs the flow of data. It fetches instructions from memory, figures out what they mean, and tells the ALU, memory, and input/output devices what to do.
- **Registers**
  - **What they are:** Extremely tiny, lightning-fast storage locations built directly *into* the CPU.
  - **Study Tip:** Think of registers as the CPU's own personal "scratchpad." The CPU uses them to temporarily hold the exact pieces of data it is actively calculating at that very millisecond.
- **Bus Structure**

- **What it is:** The electronic communication pathways built into the computer.
- **Study Tip:** Think of buses as the "highway system" of the computer. Instead of cars, these highways carry data (Data Bus), memory locations (Address Bus), and commands (Control Bus) back and forth between the CPU and other components.

- **Motherboard**
  - **What it is:** The main, large printed circuit board inside the computer case.
  - **Study Tip:** It is the "spine" or central hub of the system. Every single component—the CPU, RAM, hard drive, mouse, and keyboard—must plug into the motherboard so they can talk to each other.
- **SMPS (Switched-Mode Power Supply)**
  - **What it is:** The power box usually found at the back of a desktop computer (where you plug in the power cable).
  - **Study Tip:** Wall sockets provide AC (Alternating Current) power, but delicate computer parts need stable, low-voltage DC (Direct Current) power. The SMPS converts and regulates this electricity safely so your motherboard and CPU don't fry.
- **Expansion Slots**
  - **What they are:** Long, narrow sockets located directly on the motherboard.
  - **Study Tip:** They allow you to "expand" or upgrade your computer's capabilities in the future. For example, if you want to add a powerful graphics card for video editing or a new Wi-Fi card, you plug them into these expansion slots.